GENERATIVE AI / MODEL LIFECYCLE

How large language models are trained

From collecting training data to serving responses in production

01 THE TRAINING AND DEPLOYMENT PIPELINE

01 Collect data

Gather text and code from sources such as web pages, books, and repositories. Track source rights and provenance.

02 Prepare the dataset

Filter low-quality and sensitive content, remove duplicates, and split text into tokens (words or pieces of words).

03 Pretrain the model

Predict the next token. Compare the prediction with the actual token and update model weights to reduce error.

04 Post-train for instructions

Fine-tune on example responses. Use preference feedback to improve how the model responds.

05 Evaluate and refine

Test on held-out tasks. Check quality, bias, and safety. Use failures to guide further tuning and retesting.

06 Deploy and monitor

Serve the model through an app or API. Track response quality, latency, and safety issues after launch.

02 HOW RESOURCES SHAPE PERFORMANCE AND COST

Datasets

HOW IT HELPS

Relevant, varied examples help the model learn a wider range of tasks.

WHAT DRIVES COST

Licensing, cleaning, and labeling add cost. More tokens take more compute to process.

Compute

HOW IT HELPS

More accelerator capacity can support larger models or process more examples in the same time.

WHAT DRIVES COST

GPU rental or purchase, storage, and fast networks add cost.

Energy

HOW IT HELPS

Reliable power and cooling keep training hardware running steadily.

WHAT DRIVES COST

Electricity bills depend on energy used and local rates. Cooling and servers add to the GPU draw.

Time

HOW IT HELPS

Time for training, tuning, and tests helps teams catch and fix failures.

WHAT DRIVES COST

Longer runs tie up hardware and increase GPU-hours and staff time.

A GPU is a chip that runs many calculations in parallel. One GPU-hour means one GPU working for one hour.

NEXT / Compare actual training disclosures and cost estimates for three models on page 2.

GENERATIVE AI / TRAINING CASE STUDIES

What three models took to train

Public disclosures and cost estimates for Meta, DeepSeek, and OpenAI

META / 2024

Llama 3.1 405B

Undisclosed

Training cost in Meta’s model card

DATA

~15 trillion pretraining tokens

COMPUTE

30.84 million H100 GPU-hours
More than 16,000 H100 GPUs

ENERGY & TIME

~21.6 GWh at GPU rated power
~80 days at a constant 16,000 GPUs
GPU-only energy at 0.7 kW per chip.

DEEPSEEK / 2024

DeepSeek-V3

$5.576M

Official run at $2 per H800 GPU-hour

DATA

14.8 trillion pretraining tokens

COMPUTE

2.788 million H800 GPU-hours
Cluster of 2,048 H800 GPUs

ENERGY & TIME

Pretraining took under two months.
Energy use was undisclosed.

OPENAI / 2023

GPT-4

~$78M

Compute estimate from the 2024 AI Index

DATA

Public and licensed data sources
Token count was undisclosed.

COMPUTE

Hardware and compute totals
were undisclosed by OpenAI.

ENERGY & TIME

Energy use and training duration
were undisclosed in the technical report.

WHAT THE COST FIGURES COVER

DeepSeek’s rental estimate covers the official run and excludes earlier research and experiments.
The AI Index estimates GPT-4 compute costs using cloud rental prices. Total development budgets remain unknown.

SOURCES

Meta model card

DeepSeek-V3 technical report

GPT-4 technical report

Stanford AI Index 2024